Question	Answer	Marks	Guidance
7(a)	$G_X(t) = a + 2at + bt^2$	B1	
	$G_X'(t) = 2a + 2bt$ $E(X) = G_X'(1) = 2a + 2b$	B1	
		2	
7(b)	$G_X''(t) = 2b$	B1	
	$\text{Var}(X) = 2b + (2a + 2b) - (2a + 2b)^2$	M1	Use correct formula with *their* expressions.
	$2b + (2a + 2b) - (2a + 2b)^2 = 2b + (2a + 2b)(1 - (2a + 2b))$ $= 2b + 2(a + b)(1 - 2a - 2b)$	A1	AG, shown convincingly.
	Alternative method for question 7(b)		
	$E(X^2) = 0^2 \times a + 1^2 \times 2a + 2^2 \times b = 2a + 4b$	B1	
	$\text{Var}(X) = (2a + 4b) - (2a + 2b)^2$	M1	Use correct formula with *their* expressions.
	$(2a + 4b) - (2a + 2b)^2 = 2b + 2a + 2b - (2a + 2b)^2$ $= 2b + (2a + 2b)(1 - (2a + 2b))$ $= 2b + 2(a + b)(1 - 2a - 2b)$	A1	AG, shown convincingly.
		3	

Question	Answer	Marks	Guidance
7(c)	$\mathrm{G}_Y(t) = \left(a + 2at + bt^2\right)^{10}$	**B1 FT**	FT *their* $\mathrm{G}_X(t)$.
	$\mathrm{G}_Y'(t) = 20(a + bt)\left(a + 2at + bt^2\right)^9$ $\mathrm{E}(Y) = 20(a+b)(3a+b)^9$	**M1**	Attempt to differentiate *their* $\mathrm{G}_Y(t)$ and evaluate at $t=1$.
	$3a+b=1$, hence $\mathrm{E}(Y) = 20(a+b) = 10\mathrm{E}(X)$.	**A1**	$3a+b=1$ since total probability is 1, OE.
		3	
7(d)	Binomial	***B1**	Uses PGF to identify binomial.
	$\mathrm{B}\left(10, \frac{2}{3}\right)$	**DB1**	Identifies binomial with correct parameters.
		2	